In [21]:
```python
import pandas as pd
import numpy as np
```

In [22]:
```python
data = pd.read_csv("Data Cleaning 12.08.csv")
df = pd.DataFrame(data)
df.head()
```

Out[22]:

	name	planet_status	mass	mass_error_min	mass_error_max	mass_sini	mass
0	109 Psc b	Confirmed	5.743	0.289	1.011	6.3830	
1	112 Psc b	Confirmed	NaN	0.005	0.004	0.0330	
2	112 Psc c	Confirmed	9.866	1.781	3.190	NaN	
3	11 UMi b	Confirmed	NaN	1.100	1.100	11.0873	
4	14 And Ab	Confirmed	NaN	0.230	0.230	4.6840	

In [23]:
```python
#rows and columns
df.shape
```

Out[23]:
```
(5986, 10)
```

In [24]:
```python
#null values
df.isnull().sum()
```

Out[24]:
```
name                   0
planet_status          0
mass                3907
mass_error_min      3169
mass_error_max      3169
mass_sini           4784
mass_sini_error_min 4947
mass_sini_error_max 4947
radius              1418
radius_error_min    1893
dtype: int64
```

In [25]:
```python
#duplicated values
df.duplicated().sum()
```

Out[25]:
```
np.int64(0)
```

In [27]:
```python
# Replace infinity values (inf) with NaN (null val)
df = df.replace([np.inf, -np.inf], np.nan)

# Fix any negative values in error columns by making them positive
error_cols = [col for col in df.columns if 'error' in col]
df[error_cols] = df[error_cols].abs()
```

In [28]:
```
#fix missing mass vals - combine mass and mass_sini
df['mass'] = df['mass'].fillna(df['mass_sini'])
df.head()
```

Out[28]:

	name	planet_status	mass	mass_error_min	mass_error_max	mass_sini	ma
0	109 Psc b	Confirmed	5.7430	0.289	1.011	6.3830	
1	112 Psc b	Confirmed	0.0330	0.005	0.004	0.0330	
2	112 Psc c	Confirmed	9.8660	1.781	3.190	NaN	
3	11 UMi b	Confirmed	11.0873	1.100	1.100	11.0873	
4	14 And Ab	Confirmed	4.6840	0.230	0.230	4.6840	

In [29]:
```
df.isnull().sum()
```

Out[29]:
```
name                    0
planet_status           0
mass                 2924
mass_error_min       3169
mass_error_max       3178
mass_sini            4784
mass_sini_error_min  4947
mass_sini_error_max  4947
radius               1418
radius_error_min     1893
dtype: int64
```

In [30]:
```
#remove remaining missing
df = df.dropna(subset=['mass'])
df.isnull().sum()
```

Out[30]:
```
name                    0
planet_status           0
mass                    0
mass_error_min        245
mass_error_max        254
mass_sini            1860
mass_sini_error_min  2023
mass_sini_error_max  2023
radius               1418
radius_error_min     1500
dtype: int64
```

In [31]:
```
# Fix missing errors by fixing the empty values with 0 (otherwise calcula
error_cols = ['mass_error_min', 'mass_error_max', 'mass_sini_error_min',
for col in error_cols:
    df[col] = df[col].fillna(0)
```

In [32]:
```
df.isnull().sum()
```

Out[32]:
```
name                    0
planet_status           0
mass                    0
mass_error_min          0
mass_error_max          0
mass_sini            1860
mass_sini_error_min     0
mass_sini_error_max     0
radius               1418
radius_error_min        0
dtype: int64
```

In [34]:
```
# missing radius
df['radius'] = df['radius'].fillna(df['radius'].median())
df.isnull().sum()
```

Out[34]:
```
name                    0
planet_status           0
mass                    0
mass_error_min          0
mass_error_max          0
mass_sini            1860
mass_sini_error_min     0
mass_sini_error_max     0
radius                  0
radius_error_min        0
dtype: int64
```

In [35]:
```
# drop the redundant mass_sini column
df = df.drop(columns=['mass_sini'])
df.isnull().sum()
```

Out[35]:
```
name                 0
planet_status        0
mass                 0
mass_error_min       0
mass_error_max       0
mass_sini_error_min  0
mass_sini_error_max  0
radius               0
radius_error_min     0
dtype: int64
```

In [36]:
```
df.describe()
```

Out[36]:

	mass	mass_error_min	mass_error_max	mass_sini_error_min	mass
count	3062.000000	3062.000000	3062.000000	3062.000000	
mean	1.653834	0.298412	0.633639	0.072848	
std	2.866224	1.299681	15.478593	0.384327	
min	0.000190	0.000000	0.000000	0.000000	
25%	0.033000	0.004112	0.003592	0.000000	
50%	0.450000	0.031950	0.030000	0.000000	
75%	1.858750	0.150000	0.140000	0.006217	
max	50.000000	53.500000	850.000000	13.000000	

In [38]:

```
# Drops non-numeric columns like name
df.corr(numeric_only=True)
```

Out[38]:

	mass	mass_error_min	mass_error_max	mass_sini_erro
mass	1.000000	0.355795	0.077871	0.24
mass_error_min	0.355795	1.000000	0.052689	0.27
mass_error_max	0.077871	0.052689	1.000000	0.02
mass_sini_error_min	0.248392	0.277648	0.020508	1.00
mass_sini_error_max	0.212409	0.203593	0.031372	0.73
radius	0.184454	0.038371	0.002362	0.0
radius_error_min	-0.005694	-0.003247	-0.001227	-0.0

In [39]:

```
df['planet_status'].value_counts()
```

Out[39]:

```
planet_status
Confirmed    3062
Name: count, dtype: int64
```

In [40]:

```
# save final dataset
df.to_csv("Cleaned_Exoplanet_Data_Final.csv", index=False)
df.describe()
```

Out[40]:

	mass	mass_error_min	mass_error_max	mass_sini_error_min	mass_
count	3062.000000	3062.000000	3062.000000	3062.000000	
mean	1.653834	0.298412	0.633639	0.072848	
std	2.866224	1.299681	15.478593	0.384327	
min	0.000190	0.000000	0.000000	0.000000	
25%	0.033000	0.004112	0.003592	0.000000	
50%	0.450000	0.031950	0.030000	0.000000	
75%	1.858750	0.150000	0.140000	0.006217	
max	50.000000	53.500000	850.000000	13.000000	

In [ ]: